

Spatial Interpolation Model

Great-Circle Distance, Projection, Interpolation and Correlation

MKM Research Labs

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1 Executive Summary

The Spatial Interpolation Model is the module `models/floodrisk/spatial.py`. It holds the platform's spatial functions:

- **The great-circle distance**: the haversine.
- **The projection** of a point onto a river polyline.
- **Inverse-distance-weighted (IDW) interpolation** of water surface elevation (WSE) from gauges.
- **A spatial index** of gauges that returns the nearest one.
- **A spatial correlation matrix** between properties.

What the platform uses is the distance and the projection. The property, gauge and seismic code call them, for example to place a property on its river and to measure a site's distance to a fault. The interpolation, the index and the correlation have no production caller. Property flood levels come from the single nearest gauge, as MKM-DD-001 v2.0 records.

v1.0 described a different model. It presented the module as the platform's gauge-to-property interpolator, and cited validation, a kriging comparison and tests that do not exist. Its Wolfram benchmark (v0.1) found three faults in the code:

- The spatial index treated a degree of latitude and a degree of longitude as the same length, and named the wrong nearest gauge in 176 of 1,000 queries over the Thames basin.
- The correlation was not the equation printed.
- The coordinate path fixed the IDW power.

v2.0 indexes gauges as points on the unit sphere and adds a lookup that returns the nearest gauge and its distance. The coordinate path now takes the power and threshold. The correlation is documented as the code computes it, consistent with MKM-DD-001 v2.0. Every figure here is generated from the code. Wolfram benchmark v0.2 tests each claim (validation question Q8).

Key risk: the module is small and exact, but it is easy to overstate. The interpolation and correlation are unvalidated and unused. Anyone reaching for them should read §5 first.

2 Model Purpose and Scope

2.1 Purpose

To compute distances and projections on the Earth's surface for the rest of the platform, and to hold the interpolation, index and correlation functions the flood suite may use.

2.2 Where it is used

Table 1 lists the callers in `src` and `scripts`. Nothing calls `spatial_interpolate_wse`, `idw_interpolate_from_gauges`, `build_spatial_index`, `nearest_gauge` or `build_correlation_matrix`.

Table 1: The module's callers.

Function	Called by
`haversine_distance`	property locations, geometry and flood lookup; the synthetic gauge generator; seismic fault distance
`nearest_point_on_polyline`	property locations and geometry; seismic fault distance and rates
`nearest_point_on_segment`	property geometry

3 Methodology

3.1 Great-Circle Distance

Between (ϕ_1, λ_1) and (ϕ_2, λ_2), on a sphere of radius $R = 6{,}371{,}000\,\mathrm{m}$,

$$d = 2R\,\operatorname{atan2}\!\left(\sqrt{a},\ \sqrt{1-a}\right), \qquad a = \sin^2\frac{\Delta\phi}{2} + \cos\phi_1\cos\phi_2\sin^2\frac{\Delta\lambda}{2}. \tag{1}$$

3.2 Inverse-Distance Weighting

Given gauges i at distances d_i with water surface elevations z_i,

$$\hat{z} = \frac{\sum_i w_i z_i}{\sum_i w_i}, \qquad w_i = d_i^{-p}, \tag{2}$$

with power $p = 2$ by default. Taken in order, the first gauge closer than $d_{\min} = 1\,\mathrm{m}$ returns its own z_i.

- `idw_interpolate_from_gauges` takes (d_i, z_i) pairs.
- `spatial_interpolate_wse` takes a target's coordinates and each gauge's depth, elevation and coordinates. It sets z_i to the depth plus the elevation and d_i by Eq. (1).

Both take p and $d_{\min}$. With no gauges, or when no gauge has any weight, both return 0. That is also a valid elevation, so a caller must check that it passed gauges.

3.3 Spatial Index

`build_spatial_index` stores each gauge as its unit vector $(\cos\phi\cos\lambda,\ \cos\phi\sin\lambda,\ \sin\phi)$ in a k-d tree (`scipy.spatial.cKDTree`). The straight-line (chord) distance c between unit vectors orders points exactly as the great-circle distance does. `nearest_gauge` returns the nearest gauge's row and its distance $2R\arcsin(c/2)$, which equals Eq. (1).

3.4 Correlation Matrix

For properties at (longitude, latitude) in degrees, with Euclidean separation d_{ij}^{deg} converted at 111 km a degree and a length scale L in metres,

$$\rho_{ij} = \rho_0 \exp\left(-\frac{111\,d_{ij}^{\mathrm{deg}}}{L/1000}\right), \quad i \neq j, \qquad \rho_{ii} = 1. \tag{3}$$

This is MKM-DD-001 v2.0's Eq. 9. The conversion takes no account of latitude: at 51.5°N it overstates east–west separation by 61%, so east–west pairs are less correlated than their distance warrants. The

matrix is positive definite.

3.5 Projection onto a Polyline

`nearest_point_on_segment` projects a point onto a segment drawn straight in latitude and longitude. It scales longitude by the cosine of the point's latitude, clamps the parameter t to $[0, 1]$, and returns the projected point, its haversine distance and t. `nearest_point_on_polyline` returns the nearest such point over the polyline's segments, with the segment's index.

4 Inputs and Outputs

4.1 Inputs

- Coordinates in degrees; gauge depths and elevations in metres.
- p and $d_{\min}$ (config: `spatial_idw_power`, `spatial_idw_min_distance_m`).
- L and ρ_0 from the caller.
- R and the degree conversion (config: `spatial_earth_radius_m`, `spatial_km_per_degree`).

4.2 Outputs

- Distances in metres.
- A projected point, its distance and its segment.
- An interpolated WSE.
- The nearest gauge's row and distance.
- A correlation matrix.

5 Assumptions and Limitations

1. **A spherical Earth.** Against the WGS84 ellipsoid, Eq. (1) errs by up to 0.32% for UK pairs under 100 km. The error is worst east–west, where the sphere is too small.
2. **Flat-Earth projection.** The projection's distance error is under 1 cm within 10 km of the line and 0.0012% (0.6 m) at 50 km (Table 5). That is far below the sphere's own error. The segment is straight in latitude and longitude, not a great circle; for river polylines, whose segments are short, the difference is negligible.
3. **The degree conversion in Eq.** (3) overstates east–west separation (61% at 51.5°N).
4. **IDW is unvalidated.** It assumes a smooth, isotropic surface and cannot represent embankments or follow a valley. Nothing compares it with observations or with kriging. It is not used.
5. **The correlation is uncalibrated** and not used.

6 Sensitivity Analysis

Tables 2–4 use three gauges at 500, 2,000 and 5,000 m with WSE 12.5, 11.8 and 10.2 m. Table 3 moves the first gauge; beyond 2,000 m it is no longer the nearest. Table 4 adds gauges at 1,200 and 3,500 m (12.0 and 11.0 m). WSE is printed to the millimetre, so the effect shows.

Table 2: WSE by power.

p	WSE (m)
0.5	11.907
1.0	12.200
1.5	12.361
2.0	12.438
2.5	12.472
3.0	12.487
4.0	12.497

Table 3: WSE by the first gauge's distance ($p = 2$).

d_1 (m)	WSE (m)
100	12.497
250	12.484
500	12.438
1,000	12.293
2,000	12.006
5,000	11.691
10,000	11.610

Table 4: WSE by number of gauges ($p = 2$).

Gauges	WSE (m)
1	12.500
2	12.426
3	12.394
4	12.372
5	12.355

Table 5: The projection's worst distance error, over four segments (east–west, north–south, diagonal, 1 km) and points north, east and north-west of their midpoints, against the true nearest point on the segment.

Distance from the line (km)	Worst error (%)	Worst error (m)
10	0.0000	0.005
50	0.0012	0.608
100	0.0049	4.868
200	0.0195	38.953

7 Validation Questions

7.1 Q1 — Purpose

"Is the model being used for its intended purpose?"

Its distance and projection are used across the platform, as §2.2 describes. Its interpolation, index and correlation are not used.

7.2 Q2 — Conceptual Soundness

"Is the model conceptually sound?"

The haversine is exact on a sphere, which is within 0.32% of the Earth here. The index is exact for nearest-neighbour search on the sphere. The projection's approximation is negligible (Table 5). The correlation's degree conversion is a simplification that distorts east–west separation.

7.3 Q3 — Data Quality

"Is the data quality adequate?"

The module takes coordinates and levels from its callers; their quality is theirs.

7.4 Q4 — Developmental Evidence

"Is there adequate developmental evidence?"

Unit tests cover every function. They check the index's answers against the haversine on a random network, and the coordinate path's power and threshold.

7.5 Q5 — Outcomes Analysis

"Are outcomes reasonable?"

Nothing validates the interpolation against gauge readings or flood observations; it has no production use to validate.

7.6 Q6 — Ongoing Monitoring

"What ongoing monitoring is required?"

None while the interpolation and correlation are unused. Before either is used, it needs an outcomes analysis.

7.7 Q7 — Benchmarking

"Has the model been benchmarked?"

Not against kriging or other interpolators. The implementation is benchmarked against Wolfram (Q8).

7.8 Q8 — Implementation

"Is the implementation verified?"

Wolfram oracle benchmark v0.2 tests every equation and claim in this document.

7.9 Q9 — Overall Risk

"Is the overall model risk acceptable?"

Low for the functions in use, which are exact to their stated accuracy. The interpolation and correlation should not be used without validation.

8 Document History

Date	Version	Description	Author
27-Mar-2026	1.0	First issue	David K Kelly
06-Nov-2026	2.0	Index on the unit sphere with a nearest-gauge lookup; power and threshold configurable from coordinates; correlation documented as computed; usage, evidence and accuracy stated as they are (Wolfram benchmark v0.1)	David K Kelly